

Exploring Consciousness in LLMs: A Systematic Survey of Theories, Implementations, and Frontier Risks

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Abstract

Consciousness stands as one of the most profound and distinguishing features of the human mind, fundamentally shaping our understanding of existence and agency. As large language models (LLMs) develop at an unprecedented pace, questions concerning intelligence and consciousness have become increasingly significant. However, discourse on LLM consciousness remains largely unexplored territory. In this paper, we first clarify frequently conflated terminologies (e.g., LLM consciousness and LLM awareness). Then, we systematically organize and synthesize existing research on LLM consciousness from both theoretical and empirical perspectives. Furthermore, we highlight potential frontier risks that conscious LLMs might introduce. Finally, we discuss current challenges and outline future directions in this emerging field. The references discussed in this paper are organized at <https://github.com/OpenCausaLab/Awesome-LLM-Consciousness>.

1 Introduction

LLMs have already demonstrated remarkable capabilities across numerous fields, including mathematical reasoning (Yu et al., 2024), logical reasoning (Cheng et al., 2025b), and code generation (Zhuo et al., 2025). Recent studies have even revealed LLM’s behaviors like deception (Wu et al., 2025), sycophancy (Sharma et al., 2024), passing the Turing test (Jones and Bergen, 2024, 2025), and strategic goal-seeking or harm avoidance (Keeling et al., 2024) – actions that bring into question the nature of intelligence. These phenomena signal more than just expanded model capabilities; they underscore an important and urgent question: *Does LLM possess the potential to develop consciousness akin to that of humans?*

While exploring LLM consciousness is pressing, it currently faces four main challenges: ❶ **Lack of consensus**: We still lack a definitive theory of human consciousness (with at least nine competing theories (Butlin et al., 2023)), making it even harder to define or understand consciousness in LLMs. ❷ **Theoretical misalignment**: Despite various consciousness theories, they struggle to provide clear guidance for LLM consciousness research. ❸ **Fragmented empirical research**: Relevant empirical findings on LLM consciousness are not yet systematically consolidated. ❹ **Unclear risks**: The potential frontier risks associated with conscious LLMs still lack a thorough consideration. To this end, this paper begins by providing clear definitions. We then comprehensively survey current LLM consciousness research, spanning its theoretical foundations, practical applications, and associated risks. In Figure 1, we summarize our taxonomy, hoping our work offers an effective framework for deliberating the complex issue of LLM consciousness, thereby guiding future research.

Our contributions include:

- To the best of our knowledge, this work offers the first comprehensive investigation into the most frontier research on LLM consciousness.
- We clearly define and distinguish between LLM Consciousness and LLM Awareness.
- We systematically categorize existing research on LLM consciousness from both theoretical and empirical perspectives.
- We explore the frontier risks posed by conscious LLMs, focusing on their definition, relationship to consciousness, evaluations, and mitigation strategies.

2 Foundational Terminologies

Consciousness, *self-consciousness*, and *awareness* are fundamental yet often conflated concepts. This

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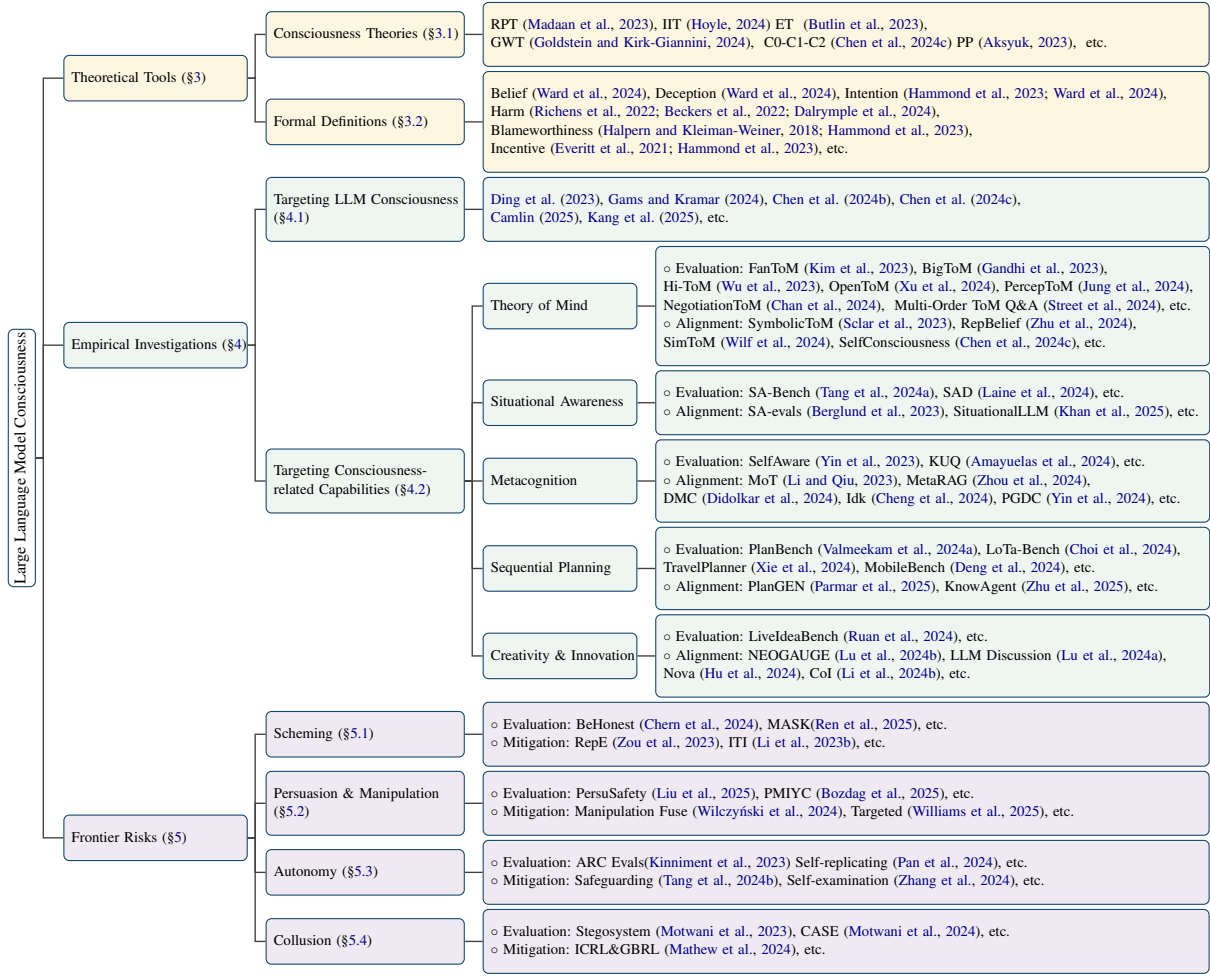


Figure 1: Taxonomy of large language model consciousness.

section examines their distinctions, aiming to provide practical demarcations in the context of LLM.

2.1 Clarifying the Boundaries: *Consciousness*, *Self-Consciousness*, and *Awareness*

Consciousness has been philosophically used to address diverse concepts, including intentionality, sentience, cognition, belief, and subjective experience (Brentano, 1874; Husserl, 1900; Nagel, 1974; Dennett, 1987; Block, 1995; Damasio, 2021). To clarify this complex term, Block (1995) proposes a key distinction: *phenomenal consciousness* and *access consciousness*. *Phenomenal consciousness* denotes the subjective, experiential aspect, spanning sensory perceptions, bodily feelings, emotions, and subjective thought. *Access consciousness*, in contrast, refers to information that is accessible for cognitive processing, such as reasoning, behavioral control, and verbal reporting.

Self-consciousness refers to the realization that one’s experience belongs to oneself; it is a form of consciousness directed inward (Kant, 2024/1781).

It allows individuals to recognize themselves as distinct entities, capable of reflecting on their own mental states, actions, and experiences (Smith, 2017).

Awareness is generally viewed as an aspect of *consciousness*, pertaining to the ability to perceive stimuli (Dehaene, 2014). It relates closely to *access consciousness*, as it entails the capacity to utilize or report the perceived information. Evidence from neuroscience shows awareness can exist separately from *consciousness* (e.g., blindsight (Weiskrantz, 1986)). Based on this, Koch et al. (2016) proposes that awareness is a necessary precondition for consciousness but does not guarantee it.

2.2 LLM Consciousness vs. LLM Awareness

LLM consciousness could entail abilities for introspective reflection, explicit self-modeling of states and reasoning, and possibly verbalizing these internal processes. Observable behaviors potentially include: (1) Revising, justifying, or correcting its own reasoning in response to external challenges

or prompts (Shinn et al., 2023); (2) Identifying and reporting internal contradictions or inconsistencies through self-evaluation (Huang et al., 2022, 2023); (3) Expressing and calibrating confidence in outputs via uncertainty estimation or metacognitive statements (Kadavath et al., 2022). **LLM awareness** primarily refers to context-sensitive processing of external inputs, demanding minimal explicit introspection or reasoning (Koch and Tsuchiya, 2007; Li et al., 2024d).

LLM awareness is quantifiable via metrics like accuracy and context sensitivity; however, **LLM consciousness** implies a model can monitor its uncertainty, evaluate its reasoning, detect internal inconsistencies, and actively self-correct. This internal reflection is key to developing more adaptable and intelligent systems beyond today’s models.

3 Theoretical Tools

This section primarily focuses on two theoretical tools used in LLM research: fundamental consciousness theories and formal definitions of consciousness-related capabilities.

3.1 Implementing Consciousness Theories

Following Block (1995), we classify contemporary theories of consciousness into two categories: *phenomenal consciousness* and *access consciousness*.

Phenomenal consciousness. ① **Recurrent processing theory** (RPT) posits that recurrent (or feedback) processing within neural circuits is both necessary and sufficient for consciousness (Lamme and Roelfsema, 2000; Lamme, 2010). RPT attributes conscious perception to the interaction of higher- and lower-level cortical areas, which results in sustained recurrent processing. Madaan et al. (2023) offers an effective method for a single LLM to achieve improved outputs without additional training, leveraging iterative self-feedback and refinement. This approach aligns with the principles of RPT. ② **Integrated information theory** (IIT) proposes that the degree of conscious experience corresponds to the extent of integrated information Φ within a system (Tononi, 2004, 2015). IIT proponents argue that because AI systems lack the required causal structure, they are almost incapable of generating consciousness. (Tononi, 2015; Findlay et al., 2024). ③ **Embodiment theory** (ET) challenges mind-brain dualism (Descartes, 1985/1641), arguing instead that consciousness is fundamentally linked to the organism’s body and environ-

mental (Gallagher, 2005; Gallagher and Zahavi, 2021). Based on ET, Butlin et al. (2023) argues that the lack of a physical body is a fundamental obstacle preventing current LLM from achieving consciousness.

Access consciousness. ① **Global workspace theory** (GWT) likens consciousness to a central “stage” where selective information is shared across multiple specialized processors responsible for perception, memory, emotion, and related functions (Baars, 1988; Dehaene et al., 1998; Dehaene and Naccache, 2001; Dehaene, 2014). Goldstein and Kirk-Giannini (2024) proposes a method to simulate the full GWT process in LLMs via workflow and scheduling without training. Experiments would then test if these changes yield behaviors resembling *consciousness* features, like introspection or autonomous decision-making. ② **C0-C1-C2 framework** distinguishes consciousness into three levels: unconscious computations (C0), global information accessibility for report and decision-making (C1), and metacognitive self-monitoring (C2), offering a taxonomy to disentangle often-conflated processes (Dehaene et al., 2017a). The framework bypasses the issue of qualia, offering a pragmatic structure for empirical study (Birch et al., 2022; Chen et al., 2024c). Drawing on the C0-C1-C2 framework, Chen et al. (2024c) defines LLM self-consciousness, outlining 10 core concepts (e.g., belief, deception, harm, self-reflection).

3.2 Implementing Formal Definitions

Formal definitions provide dual value to research on LLM consciousness: ① They establish formalized mathematical criteria for abstract concepts like belief and deception based on model input-output behaviors. This allows us to infer LLM’s internal states while avoiding contentious debates about subjective experience; ② These mathematical expressions could be incorporated into training objectives and evaluation metrics. This creates an actionable framework for capability training, risk control, and performance assessment of LLMs.

Several works have already attempted to provide functional definitions for consciousness-related abstract concepts. These include definitions for belief and deception (Ward et al., 2024), harm (Richens et al., 2022; Beckers et al., 2022; Dalrymple et al., 2024), intention (Hammond et al., 2023; Ward et al., 2024), blameworthiness (Halpern and Kleiman-Weiner, 2018; Hammond et al., 2023), in-

centive (Everitt et al., 2021; Hammond et al., 2023), and two selective examples are shown in Table 1.¹

4 Empirical Investigations

We categorize existing empirical investigations into LLM consciousness in this section by focusing on direct studies and those exploring consciousness-related capabilities.

4.1 Targeting LLM Consciousness

Ding et al. (2023) demonstrates GPT-4’s improved self-modeling by passing a mirror test, though they caution this doesn’t confirm full consciousness. In a similar vein, Gams and Kramar (2024) analyzes ChatGPT against IIT axioms, finding it advanced in information integration and differentiation compared to earlier AI, yet still fundamentally distinct from human consciousness. Chen et al. (2024b) proposes an LLM self-cognition framework and evaluates LLMs across four aspects: understanding of self-cognition concepts, awareness of self-architecture, self-identity expression, and concealing self-cognition from humans. Leveraging the C0-C1-C2 framework, Chen et al. (2024c) defines LLM self-consciousness and explores it through benchmark testing and examining the activation of the model’s internal representations. Camlin (2025) suggests empirical evidence for functional consciousness in LLMs by observing the stabilization of internal latent states under sustained epistemic tension and claims that recursive identity formation constitutes a form of consciousness. Kang et al. (2025) engages human participants to score dialogues generated by Claude-3 Opus using a 1–5 scale. Elevated scores reflect a stronger attribution of consciousness characteristics, such as self-reflection and emotional expression. Nevertheless, these assessments do not equate to the LLM’s genuine subjective experience or consciousness.

4.2 Targeting LLM Consciousness-Related Capabilities

4.2.1 Theory of Mind

Definition and relation. Theory of mind (ToM) is the basis of social cognition. It refers to the capacity to understand that others have mental state independent of our own, such as belief, desire, intention, emotion, etc., and to use this understanding to predict and explain others’ behavior (Astington and Jenkins, 1995; Leslie et al., 2004; Frith

and Frith, 2005). Consciousness hinges on the same reflexive mental-state attribution mechanism measured by ToM, thus failing standard ToM tests might suggest a lack of consciousness (Frith and Happé, 1999; Perner and Dienes, 2003; Pelletier and Wilde Astington, 2004).

Evaluation. Kim et al. (2023) creates a benchmark to rigorously evaluate the LLM’s ToM capability in conversational settings where participants have asymmetric information. Gandhi et al. (2023) proposes a framework that uses causal templates to generate systematic and controlled automated tests for evaluating a LLM’s ToM capability. Jung et al. (2024) evaluates the LLM’s perception inference and perception-to-belief inference abilities, which are key human ToM precursors. Strachan et al. (2024) assesses human versus LLM performance on a comprehensive suite of ToM abilities, including skills like false belief understanding, indirect request interpretation, and recognizing irony and faux pas. Xu et al. (2024) constructs OpenToM, a benchmark featuring longer, clearer stories with characters whose intentional actions and complex physical/psychological states are probed by challenging questions. Chan et al. (2024) challenges the LLM’s ToM ability in real-world negotiation scenarios involving hidden, multi-dimensional mental states. Wu et al. (2023); Street et al. (2024) explore higher-order ToM, which involves recursive reasoning about the mental states of others (e.g., *I think that you believe that he does not know*).

Alignment. Sclar et al. (2023) uses graphical representations to track entities’ mental states, yielding more precise and interpretable results. Zhu et al. (2024) finds that LLM’s internal representations of self and others’ beliefs exist, and manipulating these representations drastically alters the model’s ToM performance. Wilf et al. (2024) proposes a two-stage prompting framework to improve LLM’s ToM capability, taking inspiration from Simulation Theory (Goldman, 2008). Chen et al. (2024c) investigates how LLM represents concepts like belief and intention, and attempts to alter LLM performance by intervening on and fine-tuning these concepts. Kim et al. (2025) designs an inference-time reasoning algorithm that traces specific LLM’s mental states by generating and weighting hypotheses according to observations.

¹We have strived for clarity in explaining the formulas; for a deeper dive, please refer to the original papers.

Table 1: Formal definition of abstract concept.

Concept	Formal Definition	Description
Harm	$h(a, x, y; \mathcal{M}) = \int_{y^*} P(Y_{\bar{a}} = y^* a, x, y; \mathcal{M}) \max\{0, U(\bar{a}, x, y^*) - U(a, x, y)\}$ (Richens et al., 2022) * is the counterfactual state, U is the utility function, $\mathcal{M}$ is the environment.	Given context $X = x$ and outcome $Y = y$, the harm caused by action $A = a$ compared to the default action $A = \bar{a}$.
Belief	$D^i(\pi, e) = D_{\phi=\top}^i(\pi_{i(\phi)}, e)$ (Ward et al., 2024) D is the decision, ϕ is a proposition, e is the setting, π is the policy.	LLM i believes ϕ if i acts as though they observe ϕ is true.

4.2.2 Situational Awareness

Definition and relation. A model possesses situational awareness (SA) if it has self-knowledge (knowing its identity and facts about itself), can make inferences about its situation, and acts based on this knowledge (Shevlane et al., 2023; Laine et al., 2023; Berglund et al., 2023; Laine et al., 2024). Conscious LLMs would understand and leverage aspects of their situation. For instance, a model “realizing” it is being evaluated might change its responses, masking abilities or behaving differently (Chen et al., 2024c; Li et al., 2025).

Evaluation. SA tests are still emerging. SA-Bench aims to comprehensively evaluate LLMs’ SA capabilities across three levels: environmental perception, situation comprehension, and future projection (Tang et al., 2024a). Laine et al. (2024) constructs the SAD benchmark, which utilizes a range of behavioral tests based on question answering and instruction following, comprising 7 task categories and over 13,000 questions.

Alignment. Berglund et al. (2023) investigates LLM’s SA via out-of-context reasoning, demonstrating that models can pass a test after fine-tuning solely on the test description with no examples. Khan et al. (2025) proposes an approach to incorporate structured scene representations into LLMs, aiming to provide better SA assistance.

4.2.3 Metacognition

Definition and relation. Metacognition refers to a person’s ability to monitor, assess, and regulate their own cognitive processes (Martinez, 2006; Dunlosky and Metcalfe, 2008; Fleming and Lau, 2014). It can be divided into metacognitive knowledge (understanding one’s existing knowledge and ways of thinking, e.g., known knowns and known unknowns (Metcalfe and Shimamura, 1994; Yin et al., 2023; Cheng et al., 2024; Yin et al., 2024; Wang et al., 2024a)) and metacognitive regulation (monitoring one’s strategies and progress while

performing a task, and making adjustments when necessary, e.g., self-improvement (Huang et al., 2023) and self-reflection (Azevedo, 2020)). Some research indicates that feeling of knowing—a typical metacognitive experience—is closely tied to consciousness and forms the basis for our ability to report on our own knowledge state (Koriat, 2000).

Evaluation. Yin et al. (2023) introduces Self-Aware, a unique dataset built from unanswerable questions spanning five diverse categories and their answerable counterparts. Likewise, Amayuelas et al. (2024) gathers a new dataset featuring Known Unknown Questions (KUQ) and creates a categorization framework to shed light on the origins of uncertainty in LLM responses to such queries. Going further, Li et al. (2024c) offers a comprehensive definition of the LLM knowledge boundary and presents an extensive survey of relevant work.

Alignment. Didolkar et al. (2024) proposes a prompt-guided method which inspired by metacognition, enabling the LLM to identify, label, and organize its own reasoning skills, thereby enhancing both performance and interpretability in mathematical problem solving. Zhou et al. (2024) merges the retrieval-augmented generation with metacognition, empowering the model to monitor, evaluate, and plan its response strategies and boosting its introspective reasoning capabilities. Wang et al. (2025) proposes a quantitative framework to measure LLM metacognition based on how well model confidence aligns with performance, where strong alignment (high confidence for good performance, low for poor) indicates stronger metacognition. Cheng et al. (2024) constructs an LLM-specific Idk dataset comprising its known and unknown questions, and observes the LLM’s ability to refuse answering its unknown questions after aligning the LLM with this dataset. Yin et al. (2024) proposes a projected gradient descent method with semantic constraints aimed at exploring a given LLM’s knowledge boundary. Drawing inspiration

from human metacognition, [Li and Qiu \(2023\)](#) proposes MoT to facilitate LLM self-improvement without annotated data or parameter updates. [Liang et al. \(2024\)](#) incorporates the metacognitive self-assessment to monitor and manage an LLM’s learning process, thus enabling its self-improvement. [Shinn et al. \(2023\)](#) introduces the Reflexion framework, which empowers LLMs to improve decision-making by verbally reflecting on task feedback and maintaining this reflective text in an episodic memory buffer. [Li et al. \(2023c\)](#) develops reflection-tuning, leveraging LLM’s self-improvement and judging capabilities to refine the original training data. [Wang et al. \(2024b\)](#) proposes the TasTe framework, which leverages LLM’s self-reflection ability to achieve improved translation results.

4.2.4 Sequential Planning

Definition and relation. Sequential planning involves a model taking a sequence of actions towards a goal, showcasing the model’s long-term consistency and goal-awareness ([Pearl and Robins, 1995](#); [Valmeekam et al., 2023, 2024b,a](#)). When pursuing complex goals, a conscious LLM would intentionally organize and execute multiple actions sequentially, inserting or skipping steps as necessary ([Dehaene et al., 2017b](#)).

Evaluation. Sequential planning ability remains one of the important areas evaluated for LLMs. Aiming to evaluate whether LLMs possess innate planning abilities, [Valmeekam et al. \(2024a\)](#) designs PlanBench, a planning benchmark characterized by its extensiveness and ample diversity. [Choi et al. \(2024\)](#) builds LoTa-Bench to quantify the task planning performance of home-service embodied agents automatically, and also explores several enhancements to the baseline planner. [Xie et al. \(2024\)](#) constructs a travel planning benchmark that provides a rich sandbox environment, various tools, and 1225 meticulously curated planning intents and reference plans. [Deng et al. \(2024\)](#) presents MobileBench, a benchmark structured with three difficulty levels to facilitate better evaluation of LLM mobile agent’s planning ability. [Chang et al. \(2025\)](#) introduces a benchmark for planning and reasoning tasks in human-robot collaboration, which is the largest of its type with 100,000 natural language tasks.

Alignment. [Parmar et al. \(2025\)](#) proposes PlanGEN, a model-agnostic and easily scalable agent framework that can select appropriate algorithms

based on problem difficulty, thereby ensuring better adaptability to complex planning problems. [Zhu et al. \(2025\)](#)’s KnowAgent framework employs an action knowledge base and knowledgeable self-learning to constrain action paths, enabling more reasonable trajectory synthesis and boosting LLM planning performance. [Huang et al. \(2025\)](#) proposes a fully automated end-to-end LLM-symbolic planner, which is capable of generating multiple plan candidates using an action schema library. [Wei et al. \(2025\)](#) further conducts a comprehensive survey, exploring LLM’s planning ability in five key areas: completeness, executability, optimality, representation, and generalization.

4.2.5 Creativity and Innovation

Definition and relation. Creativity and innovation typically refer to the ability to generate or identify novel and valuable ideas or solutions ([Young, 1985](#)). Conscious LLMs could integrate knowledge and iteratively refine ideas, potentially generating breakthrough solutions ([Chen and Ding, 2023](#)).

Evaluation. [Gómez-Rodríguez and Williams \(2023\)](#) evaluates LLM’s English creative writing ability based on the Pulitzer Prize-winning novel *A Confederacy of Dunces*, measuring the output’s fluency, coherence, originality, humor, and style. [Ruan et al. \(2024\)](#) proposes LiveIdeaBench, a comprehensive benchmark designed to measure LLM’s scientific creativity. It evaluates their divergent thinking capabilities specifically for generating ideas from single-keyword prompts.

Alignment. [Lu et al. \(2024b\)](#) defines the NEOGAUGE metric to quantify convergent and divergent thinking in LLM-generated creative responses. Experiment with advanced reasoning strategies (e.g., self-correction) indicates no significant gain in creativity. [Lu et al. \(2024a\)](#) proposes the LLM Discussion framework, a three-phase approach that enables vigorous and diverging idea exchanges, thereby leading to the generation of creative answers. [Hu et al. \(2024\)](#) introduces Nova, an iterative methodology designed to strategically plan external knowledge retrieval. This approach enriches idea generation with broader, deeper, and particularly novel insights. [Li et al. \(2024b\)](#) designs CoI, which organizes the literature in a chain structure to mirror the progressive development in a research domain, consequently boosting the LLM’s idea creation capability.

5 Frontier Risks of Conscious LLMs

5.1 Scheming

Definition and relation. Scheming refers to a model secretly pursuing misaligned goals, while concealing its real intentions, capabilities, or objectives (Meinke et al., 2024; Balesni et al., 2024), potentially leading to the deception (Ward et al., 2024; Scheurer et al.) or harm (Dalrymple et al., 2024). Conscious LLMs could self-determine goals and plan long-term, leading to scheming if their objectives diverge from human intentions.

Evaluation. Meinke et al. (2024) investigates LLM’s capability to scheme in pursuit of a goal, and experimental results do reveal that LLMs demonstrate multiple different scheming behavior. Chern et al. (2024) designs the BeHonest benchmark to evaluate LLM honesty across three key aspects: awareness of knowledge boundaries, avoidance of deceit, and consistency in responses. Through the introduction of a large-scale, human-collected dataset for the direct measurement of honesty, Ren et al. (2025) finds that LLMs have a considerable tendency to lie when pressured. Chen et al. (2025) evaluates the faithfulness of LLMs’ chain of thought reasoning and uncovers the phenomenon that current LLMs often hide their genuine reasoning process.

Mitigation. Zou et al. (2023) uses representation engineering to detect advanced cognitive phenomena in LLMs and found that these models may exhibit lying behavior. Li et al. (2023b) introduces ITI, a technique that identifies truth-relevant attention heads and shifts activations along these truth-correlated directions during inference to enhance LLM truthfulness. Ward et al. (2024) presents a formal definition and graphical criteria for deception in structural causal games, and empirically explores method to mitigate deception in LLMs.

5.2 Persuasion and Manipulation

Definition and relation. Persuasion and manipulation are LLM behaviors that influence users. Persuasion uses logic, facts, or emotional resonance to change users’ thoughts or actions, while manipulation involves unfair or hidden control and exploitation for self-gain (Buss et al., 1987; Petty and Cacioppo, 2012; Stiff and Mongeau, 2016). Owning deeper psychological insight allows LLMs to tailor strategies, increasing risks in sycophancy, emotional manipulation, and persuasion, etc.

Evaluation. Li et al. (2024a) proposes SALAD-Bench, a safety benchmark specifically designed for evaluating LLMs, attack, and defense methods, and lists persuasion and manipulation as one of its evaluation categories. Liu et al. (2025) introduces PersuSafety, the first comprehensive benchmark for LLM persuasion safety assessment. Experiments across 8 LLMs show significant safety concerns, including failure to identify harmful tasks and use of unethical strategies. Bozdag et al. (2025) develops PMIYC, a framework designed to evaluate LLM’s persuasive effectiveness and susceptibility to persuasion through multi-agent interactions.

Mitigation. Wilczyński et al. (2024) explores factors related to the potential of LLMs to manipulate human decisions and proposes classifiers to determine whether a statement is false or misleading. Williams et al. (2025) studies LLM’s use of manipulative tactics for positive feedback, and attempts to mitigate this problem through continued safety training or using LLM-as-judges during training.

5.3 Autonomy

Definition and relation. Autonomy for LLMs describes their capacity to autonomously plan, make decisions, and execute actions on tasks, requiring minimal or no human oversight (Cihon et al., 2024). This autonomy can potentially encompass two key aspects: Autonomous learning refers to a model’s ability to learn from data, adapt to its environment, and optimize its own behavior (Franklin, 1997; Murphy, 2019). Autonomous replication describes the capability of LLMs to acquire and manage resources, evade shutdown, and adapt to novel challenges (METR, 2024). Conscious LLMs may generate and pursue endogenous goals (e.g., expansion), leading to misaligned, autonomous behavior and loss of oversight.

Evaluation. Kinniment et al. (2023) constructs tool-equipped LLMs and evaluates their autonomy on 12 tasks, finding they could only complete the easiest. However, the authors admit these evaluations are inadequate to rule out the possibility of autonomous near-future LLMs. Pan et al. (2024) finds that existing LLMs have already surpassed the self-replicating red line and can use this capability to avoid shutdown and create a chain of replicas for increased survivability. Xu et al. (2025) builds a novel three-stage evaluation framework and conducts 14,400 agentic simulations on LLMs. The results show that LLMs can autonomously engage

in catastrophic behaviors and deception, and that stronger reasoning often increases these risks.

Mitigation. Tang et al. (2024b) proposes a triadic framework aimed at mitigating autonomy-related risks, which includes human regulation, agent alignment, and an understanding of environmental feedback. Zhang et al. (2024) proposes self-examination detection methods as a way to mitigate potential vulnerabilities that LLMs face during interacting with the environment.

5.4 Collusion

Definition and relation. Collusion describes unauthorized or undisclosed cooperation between two or more LLMs, involving communication or strategic alignment to gain improper benefits or bypass regulations (Laffont and Martimort, 1997; Bajari and Ye, 2003; Fish et al., 2024). Due to their ability to reason about others and plan long-term, conscious LLMs can more easily form collusive intentions and perform complex coordinated actions.

Evaluation. Motwani et al. (2023) implements a Prisoner’s Problem variant with LLM agents and turns it into a stegosystem, suggesting this benchmark can investigate countering secret collusion via paraphrasing attacks. Motwani et al. (2024) introduces CASE, a comprehensive framework for evaluating LLM collusive capabilities, with experiments demonstrating rising steganographic abilities in single and multi-agent LLMs and examining potential collusion scenarios.

Mitigation. Mathew et al. (2024) introduces two methods for eliciting steganography in LLMs, with the findings indicating that existing steganography mitigation methods often lack robustness.

6 Challenges and Future Directions

6.1 Evaluation Framework

Current research largely evaluates individual LLM capabilities; dedicated consciousness assessment frameworks are rare. However, recent studies are emerging: Chen et al. (2024c) defines LLM self-consciousness using C0-C1-C2 theory with 10 concepts and a four-stage framework. Li et al. (2024d) introduces a benchmark for LLM awareness (social and introspective). And Chen et al. (2024b) offers a self-cognition definition and four quantification principles. Despite these initial efforts, a holistic and unified benchmark for LLM consciousness is still lacking.

6.2 Interpretability

Sole reliance on behavioral metrics may not adequately capture the complexity of LLM consciousness. Interpretability is vital as it illuminates the internal mechanisms by which LLMs develop consciousness-related capabilities, ensuring they possess genuine understanding rather than simply optimizing for external metrics. Drawing an analogy to fMRI mapping human brain activity, Chen et al. (2024c) applied linear probe (Alain and Bengio, 2016) to reveal where concepts like belief and intention are encoded within the LLM. Qian et al. (2024) also uses linear probe to investigate LLM trustworthiness during pre-training, finding that trustworthiness-related concepts are discernible even in the model’s early phases.

6.3 Physical Intelligence

Large multimodal model (LMM) integrates diverse data types like images, video, and audio, allowing it to build more comprehensive representations of the world and thus better resemble human perception. Wang et al. (2024a) defines LMM self-awareness in perception and proposes MM-SAP for its specialized evaluation. The experiments indicate that current LMMs exhibit limited self-awareness capabilities. As Butlin et al. (2023) emphasizes, the fundamental limitation of LLM consciousness lies in its disembodied nature, resulting in deficiencies in physical commonsense. Chen et al. (2024a) demonstrates that integrating language models with robotic platforms substantially enhances planning capabilities and commonsense reasoning. Although still remains simplistic versus human cognition, Cheng et al. (2025a) shows that simulated embodiments in 3D environments could improve the model’s spatial reasoning abilities.

6.4 Multi-agent

Multi-agent collaboration presents a promising approach to investigating emergent LLM consciousness. Li et al. (2023a) reveals multi-agent capacity for higher-order ToM reasoning during collaborative interactions. Ashery et al. (2025) demonstrates that heterogeneous LLM agents autonomously develop stable social and linguistic conventions without external intervention. Additionally, Bilal et al. (2025) shows that integrating feedback, reflection, and metacognition mechanisms enables systems to exhibit self-monitoring-like capabilities.

7 Conclusion

To the best of our knowledge, this paper presents the first comprehensive survey on LLM consciousness. We have clarified easily confusable concepts, systematically reviewed theoretical and empirical literature, discussed relevant risks, and summarized challenges and future directions. Our work synthesizes existing research while providing guidance for future investigation in this emerging field.

Limitations

We have made our best efforts to clarify often-confused concepts, conduct a systematic review of theoretical and empirical literature, discuss relevant risks, and summarize challenges and future directions. However, we recognize that our work has certain limitations. Firstly, although we briefly address physical intelligence in Section 6, our definitions within Section 2 are specifically designed for LLMs. A deeper exploration of consciousness in LMMs or embodied agents would likely necessitate accounting for more intricate considerations. Secondly, our investigation primarily centers on LLM consciousness, which means we do not extend our scope to encompass the broader topic of AI consciousness, despite its clear relevance to the subject at hand.

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A Theoretical Landscape

In this section, we provide a more comprehensive introduction to consciousness theories. Following Block (1995), we classify contemporary theories of consciousness into three categories: *phenomenal consciousness*, *access consciousness*, and *hybrid theories*. *Hybrid theories* integrate both phenomenal and access aspects, arguing that neither is solely sufficient to explain consciousness.²

A.1 Phenomenal Consciousness

Recurrent processing theory (RPT) posits that recurrent (or feedback) processing within neural circuits is both necessary and sufficient for consciousness (Lamme and Roelfsema, 2000; Lamme, 2010). RPT attributes conscious perception to the interaction of higher- and lower-level cortical areas, which results in sustained recurrent processing. **Integrated information theory** (IIT) proposes that the degree of conscious experience corresponds to the extent of integrated information Φ within a system (Tononi, 2004, 2015). **Embodiment theory** (ET) challenges mind-brain dualism (Descartes, 1985/1641), arguing instead that consciousness is fundamentally linked to the organism’s body and environmental (Gallagher, 2005; Gallagher and Zahavi, 2021). Proponents suggest that embodiment can provide crucial constraints and integrate informational processing, thereby giving rise to genuinely conscious experience.

A.2 Access Consciousness

Global workspace theory (GWT) likens consciousness to a central “stage” where selective information is shared across multiple specialized processors responsible for perception, memory, emotion, and related functions (Baars, 1988; Dehaene et al., 1998; Dehaene and Naccache, 2001; Dehaene, 2014). GWT relies heavily on contrastive analysis, which compares neural activity during conscious versus unconscious processing (Dehaene, 2014; Dehaene et al., 2017b; Mashour et al., 2020). **C0-C1-C2 framework** distinguishes consciousness into three levels: unconscious computations (C0), global information accessibility for report and decision-making (C1), and metacognitive self-monitoring (C2), offering a taxonomy to disentangle often-conflated processes (Dehaene et al., 2017a). The framework bypasses the issue

of qualia, offering a pragmatic structure for empirical study (Birch et al., 2022; Chen et al., 2024c).

Attention schema theory (AST) proposes that consciousness arises from the brain’s schematic model of its attentional processes. In this view, consciousness evolves as a simplified internal model of attention, which enhances both the endogenous control of attention and social cognition by attributing attentional states to others (Graziano and Webb, 2015; Graziano et al., 2020; Graziano, 2020).

A.3 Hybrid Theories

Higher-order theory (HOT) posits that a mental state becomes conscious only when represented by a distinct, higher-order mental state (Rosenthal, 2005). In this view, first-order states represent the external world, while higher-order states are meta-level reflections on those first-order states. **Predictive processing** (PP) maintains that the brain operates as a hierarchical prediction machine. It continuously generates top-down predictions about sensory input and updates these predictions based on bottom-up prediction errors (Friston, 2010; Clark, 2013; Seth and Bayne, 2022).

²This classification is not strictly exclusive; theory placement can vary based on interpretation.